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Project Enrollment
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A01 Enrollment

Business Rules:

1. Every student has their own student ID uniquely identifying their information such as full name, address, phone number, and email. A student can major in one or more majors. A major cannot be deleted while other students are still in it. Deleting a student removes all of their enrollments.
2. Every course has its own course ID, which identifies its course information such as name, units, department. A course can be offered to different majors. A major can offer multiple courses as well.
3. A course has multiple sections. Each section has its own section number, which is unique within its own course, but sections from different courses can have the same section number. A section number may repeat over each term. A section has a term, start and end date, capacity, waitlist capacity, weekly schedule, location. Each section can only be taught by one instructor. A section can be temporarily instructorless.
4. A student enrolled in a section gets a grade at the end of the semester. A student cannot enroll in the same section twice. A student cannot enroll in the same course in the same term twice.
5. An instructor can be identified with an instructor_id, who has a first and last name. An instructor can teach multiple sections across different courses. An instructor cannot simultaneously teach 2 sections.
6. Some courses require other courses as prerequisites before enrollment. These prerequisites must be passed with a certain grade to be cleared. Some prerequisites can be substituted by transfer courses or AP credit.
7. Sections cannot be enrolled once max capacity is reached. Instead students are enrolled into the waitlist. If space opens in the section, students are automatically enrolled into the section. If the waitlist is at max capacity, reject any enrollments. Waitlist priority is by position. A student can not be enrolled and waiting at the same time.
8. Students have a max of 18 units per semester. Students who try to enroll past this cap fail to enroll. Students can get a unit cap raise / override for the semester, up to 21 units, approved by an instructor, advisor, department chair, or dean.
9. Students cannot enroll in conflicting section schedules.
10. If a course is deleted, its sections and their enrollments and waitlists are deleted too. If an instructor is deleted, their sections go back to TBA.

SQL Problem:

How should the database handle automatic enrollment into a section from the waitlist when a space opens up?

1. Which business rule is this about?

Business Rule #7 - Sections cannot be enrolled once max capacity is reached. Instead students are enrolled into the waitlist. **If space opens in the section, students are automatically enrolled into the section.** If the waitlist is at max capacity, reject any enrollments. Waitlist priority is by position. A student can not be enrolled and waiting at the same time.

2. Is that rule on your ER, with its cardinality and participation?

No.

3. Which relation and key did the relational model give you for it?

N/A, we designated this to be a trigger problem.

4. Does the rule look at one row, a key, another table, or many rows?

Many rows.

5. Only then: what does the Java side show, and what does the server side show?

At this stage of the project, our Java is not yet functional.